

COMPUTER SCIENCE IN CONTEXT

As you can see from studying problem statements, many systems use more than one sensor to monitor processes such as manufacturing. The data from each of these sensors provides an input into a logic circuit to determine what the output will be. Some manufacturing business use as many as ten different sensors in a logic circuit to monitor their manufacturing processes. That would be one huge logic circuit!

SUMMARY

A logic gate is a component in a computer that controls the flow of electricity.

A single value is output from a logic gate, this can be a 1 or 0. 1 sets it to high, 0 sets it to low.

A problem statement is a description of the conditions of an electrical system.

A logic circuit is a diagrammatic representation of an electrical system showing the logic gates it uses. The logic gates it can use are NOT, AND, OR, NAND, NOR and XOR.

A truth table is a representation of all the different outputs of an electrical system, dependent on the values of each input.

A logic expression is a mathematical representation of an electrical system.

EXAM-STYLE QUESTIONS

- 1 **State** the purpose of a logic gate in a computer system. [1]
- 2 Draw the single logic gate that would generate the given truth tables:

a

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

[1]

b

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

[1]

[Total: 2]

COMMAND WORD

state: express in clear terms.

CONTINUED

3 Consider the truth table:

A	B	X
0	0	1
0	1	0
1	0	1
1	1	1

a Write a logic expression to represent the given truth table. [4]

b Draw a logic circuit to represent the given truth table. [4]

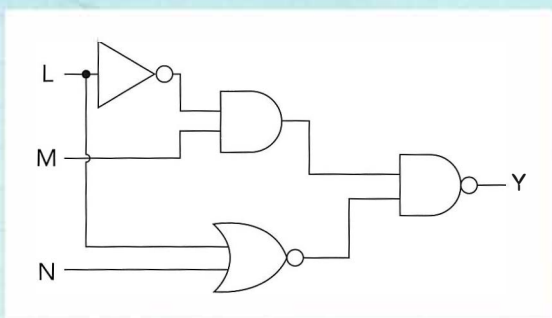
[Total: 8]

4 Consider the truth table:

S	T	P	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Write a logic expression to represent the given truth table. [5]

5 Consider the logic circuit:



a Write a logic expression to represent the given logic circuit. [3]

CONTINUED

- b Complete the truth table to represent the given logic circuit:

L	M	N	Workspace	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

[Total: 7]

- 6 A greenhouse has a system that monitors the growing conditions for fruits and vegetables. The system has three conditions to monitor the moisture level of the soil (M) the temperature (T) and the humidity (H). The conditions for the system are:

Input	Binary value	Condition
M	1	Moisture level $\leq 80\%$
	0	Moisture level $> 80\%$
T	1	Temperature $> 35^\circ\text{C}$
	0	Temperature $\leq 35^\circ\text{C}$
H	1	Humidity $\geq 65\%$
	0	Humidity $< 65\%$

The system will sound an alarm (X) when certain conditions are detected.
The alarm will sound when:

The moisture level $\leq 80\%$ and the humidity $< 65\%$

or

The temperature $> 35^\circ\text{C}$ and the humidity is $\geq 65\%$

- a Write a logic expression to represent the problem statement for the alarm system for the greenhouse. [3]
- b Draw a logic circuit to represent the problem statement for the alarm system for the greenhouse. [5]
- c Complete the truth table to represent the problem statement for the alarm system for the greenhouse.

M	T	H	Workspace	X
0	0	0		
0	0	1		

CONTINUED

M	T	H	Workspace	X
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

[Total: 12]

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics. This will help you to see any gaps in your knowledge and help you to learn more effectively.

You might find it helpful to rate how confident you are for each of these statements when you are revising. You should revisit any topics that you rated 'Needs more work' or 'Getting there'.

I can...	See topic	Needs more work	Getting there	Confident to move on
draw logic gates using the standard symbols.	10.1–10.7			
describe the logic of each different logic gate.	10.1–10.7			
complete a truth table for each logic gate.	10.1–10.7			
identify the single logic gate the represents a given truth table.	10.1–10.7			
complete a truth table to represent a logic circuit.	10.1–10.7			
write a logic expression to represent a logic circuit.	10.8			
write a logic expression to represent a truth table.	10.9			
draw a logic circuit to represent a truth table.	10.9			
draw a logic circuit to represent a logic expression.	10.10			
complete a truth table to represent a logic expression.	10.10			
write a logic expression to represent a problem statement.	10.11			
draw a logic circuit to represent a problem statement.	10.11			
complete a truth table to represent a problem statement.	10.11			